Veteran Status and Earnings: Evidence from IPUMS Data

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Introduction

This paper studies the effect of veteran status on wage income among those who hold a bachelor's degree. We have a total of $500,000$ observations from the IPUMS ACS, from 2010 to 2019. We include control variables for age, disability rating, physical and cognitive difficulty, sex, degree field, occupation, whether they live in a metropolitan area, and survey year fixed effects. The specific question of interest is: *What is the effect of veteran status on wage income among those who hold a bachelor's degree?*.

This question matters because military service may affect later labor market outcomes in several ways. On one hand, veterans may gain skills, such as discipline, leadership, and even valuable technical work experience that improves wages in the civilian market. On the other hand, military service may interrupt career development or cause health limitations that reduce earnings or the ability to work certain jobs, which could reduce earnings. Understanding this relationship is relevant for evaluating the transition of veterans into the civilian market and labor outcomes.

Estimating this effect is difficult because veteran status is not randomly assigned. Veterans may differ from non-veterans in ways that could also affect earnings, such as age, gender, geography, and labor market experience. These factors may confound the relationship between veteran status and wages, so the raw differences in average income cannot automatically be interpreted as causal effects.

This report presents a preliminary descriptive analysis of the data set. The goal is to understand the main patterns in the data and identify potential confounding factors.

Data

The data used in this project came from the Integrated Public Use Microdata (IPUMS), a harmonized source of U.S. Census and American Community Survey microdata. IPUMS provides individual-level information on demographic characteristics, education, earnings, and occupation. The project uses annual observations from multiple years and focuses on individuals between the ages of 25 and 64 with a bachelor's degree.

The key variables used in the analysis are wage income **inwage**, veteran status **VETSTAT**, age **AGE**, sex **SEX**, education level detailed **EDUCD**, degree field **DEGFIELD**, occupation **OCC2010**, cognitive difficulty **DIFFREM**, physical difficulty taking care of themselves **DIFFCARE**, VA service-connected disability rating **VETDISAB** and the survey year **YEAR**. Wage income is measured as total annual wage and salary income, excluding VA benefits, pension, or disability compensation.Veteran status is coded as an indicator variable equal to 1 for veterans and 0 for non-veterans. Because wage income is highly skewed, the analysis uses the natural logarithm of wage income in figures and regressions. The unit of observation is an individual in a given year, so the data set is organized at the person-year level. The final sample consists of $500,000$ observations after applying all sample restrictions..

Descriptive Analysis

We begin with an overall summary of the main variables in the sample. Table 1 provides a broad picture of the population under study and establishes the scale of wages, age, and gender composition before comparing veterans and non-veterans.

Table 1: Overall Summary Statistics

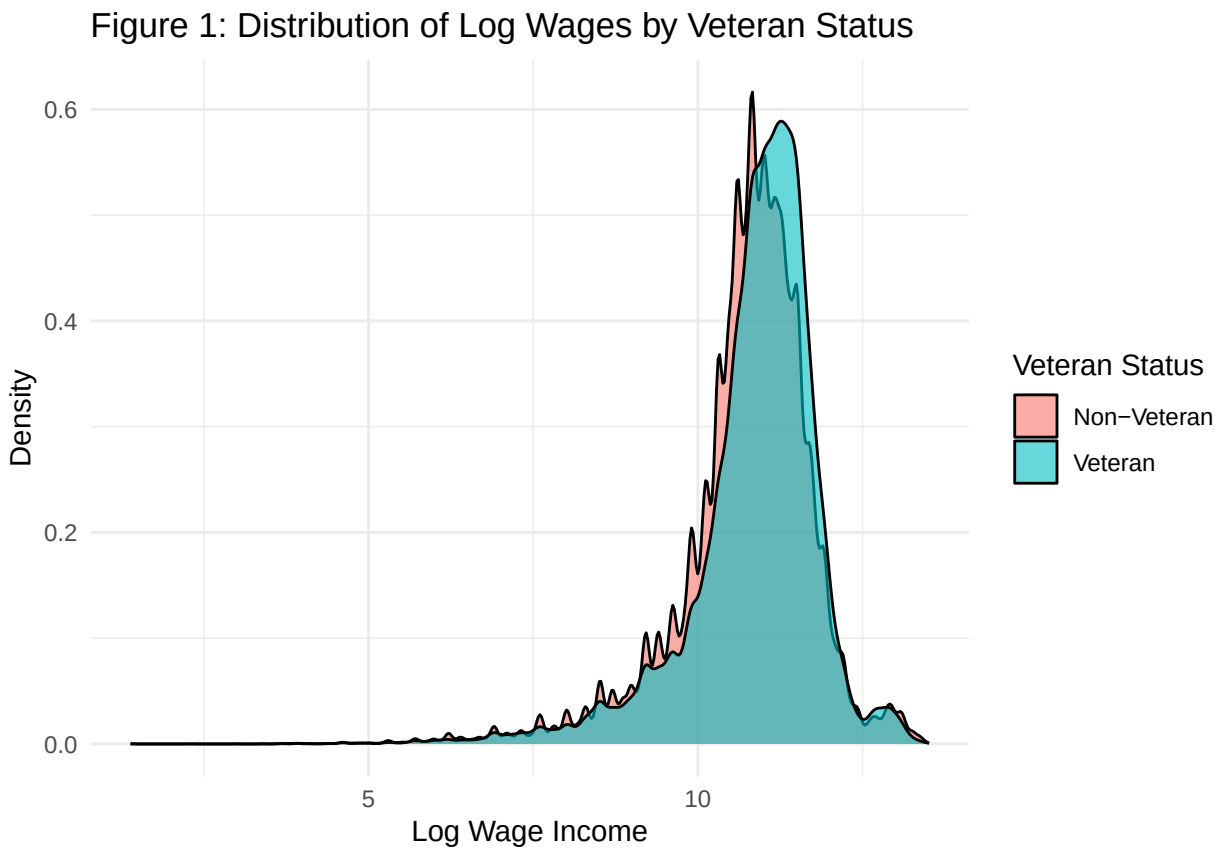
N	mean_wage	median_wage	mean_log_wage	mean_age	female_share	veteran_share
500,000	68105.05	52000	10.72532	42.89969	0.51171	0.047028

Table 1 reports summary statistics for the full sample. It shows the average and median wage, the average log wage, average age, and the shares of women and veterans in the sample. These are useful because they describe the basic composition of the dataset before turning to differences by veteran status.

Table 2: Summary Statistics by Veteran Status

veteran	N	mean_wage	median_wage	mean_log_wage	mean_age	female_share
0	476,486	67811.88	51000	10.71868	42.64063	0.5288802
1	23,514	74045.82	60000	10.85985	48.14932	0.1637748

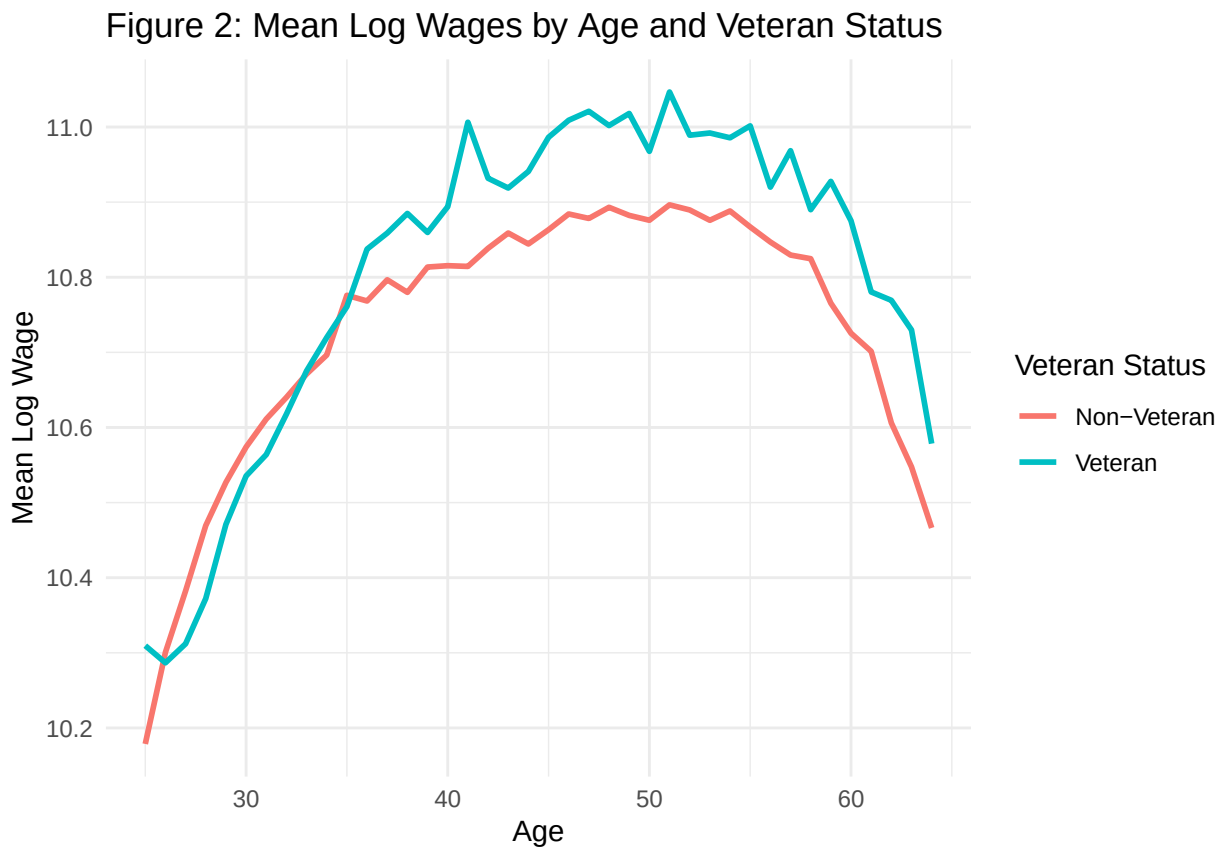
Table 2 compares veterans and non-veterans directly. In the raw data, veterans appear to earn more on average than non-veterans. However, veterans are also older on average and much less likely to be female. These differences are important because both age and gender are strongly related to wages. As a result, the raw wage gap is not necessarily informative about the causal effect of veteran status.



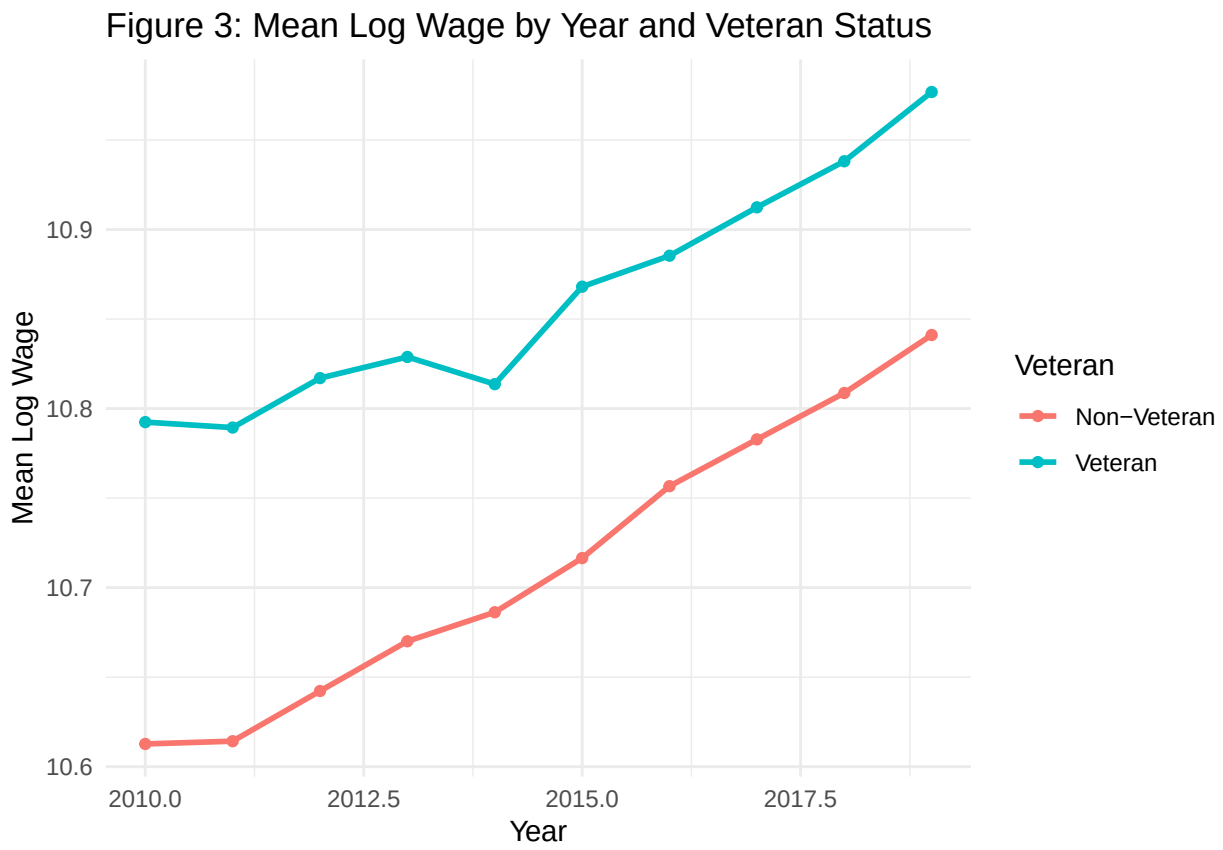
In **figure 1**, we can see the distribution of log wage income for veterans and non-veterans. The distribution for veterans is shifted to the right compared to non-veterans, indicating that veterans tend to have higher log wage income on average. However, there is also a wider spread in the distribution for veterans, which could be due to the fact that veterans have a wider range of experiences and backgrounds compared to non-veterans. Additionally, there is some overlap between the two distributions, suggesting that there are some non-veterans who earn more than some veterans, and vice versa.

Let's look at another figure; figure 2, shows the distribution of mean log wages by age and veteran status. This provides a visual representation of how wages progress with age.

Because age is likely a confounder, it is useful to examine how wages vary with age for veterans and non-veterans. This is going to help assess whether older average age among veterans may explain part of the raw wage gap we see.



In **figure 2**, we can see that mean log wage income increases with age for both veterans and non-veterans. However, at younger ages, veterans are earning slightly less than non-veterans, but around the age of 33-35, veterans start to earn consistently more than non-veterans on average. One possible explanation is differences in career experience, though this cannot be directly tested with the data. Additionally, we can look at the graph of mean log wage by year and veteran status to see if there are any trends over time, and if they are consistent with the overall trends seen in figures 1 and 2.



In **figure 3**, mean log wage increases consistently over time for both veterans and non-veterans, but the gap between the two groups remains relatively constant over time with veterans earning more on average than non-veterans. This suggest that the relationship between veteran status and wage income is not being driven by any specific year, and that the trends we are seeing in figures 1, 2, and 3 are consistent over time.

Economic Model

To estimate the effect of veteran status on wage income, we use the following regression model, slowly adding more controls to see how the veteran coefficient changes. The dependent variable is the natural logarithm of wage income, and the key independent variable is veteran status. The model is as follows:

$$\log(wage_{it}) = \beta_1 Veteran_{it} + X'_{it}\gamma + \delta_t + \theta_j + \omega_m + \epsilon_{it}$$

Where X_{it} controls age, sex, cognitive and physical difficulty, and disability rating. δ_t are year fixed effects, θ_j are degree field and occupation fixed effects, and ω_m are metropolitan area fixed effects.

Results

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Table 3: Regression Results

	Dependent variable:				
	Log Wage Income				
	OLS		*felm*		
	Raw	Controls	METRO + Year	Degree + Job	Full Model
	(1)	(2)	(3)	(4)	(5)
Veteran	0.141***	−0.092***	−0.071***	−0.066***	−0.023***
	(0.007)	(0.007)	(0.007)	(0.006)	(0.008)
Female		0.010***	0.010***	0.008***	0.008***
		(0.0001)	(0.0001)	(0.0001)	(0.0001)
Age		−0.493***	−0.486***	−0.295***	−0.295***
		(0.003)	(0.003)	(0.003)	(0.003)
DIFFCARE					−0.318***
					(0.033)
DIFFREM					−0.462***
					(0.017)
VETDISAB					−0.019***
					(0.003)
Constant	10.719***	10.565***			
	(0.002)	(0.006)			
Observations	500,000	500,000	500,000	500,000	500,000
R^2	0.001	0.068	0.083	0.292	0.295
Adjusted R^2	0.001	0.068	0.083	0.291	0.294
Note:					*p<0.1; **p<0.05; ***p<0.01

In regression (1), we see that veterans earn 14.1% more than non-veterans on average. However, the raw wage premium for veterans disappears once controls are added, there is a significant decrease in the veteran coefficient. In regression (2), when adding controls for age and gender, veterans earn 9.2% less than non-veterans on average. As we start adding fixed effects for year and metropolitan area in regression (3), the gap starts to close, and veterans earn 7.1% less than non-veterans on average. Year fixed effects control for macroeconomic conditions, while occupation and degree fixed effects control for differences in job type and education. In regression (4), after adding degree field and occupation fixed effects, the gap decreases to 6.6%. Finally, in regression (5), after adding controls for cognitive difficulty, physical difficulty, and VA disability rating, the gap decreases to 2.3% less than non-veterans on average. However, because disability may itself be an outcome of military service, controlling for it may understate the total effect of veteran status on wages.

As a result, this estimate should be interpreted as conditional rather than the total effect. The reversal of the veteran coefficient from positive to negative suggest strong selection effects, where veterans differ systematically from non-veterans. An interesting observation about these regressions is the increase of R^2 as we add more controls, especially in regression (4) and (5), which suggest that degree field, occupation, and disability are important factors in explaining wage income, and the observations are getting closer to the regression line as we add more controls.

However, these results should not be interpreted as causal, as veteran status is not randomly assigned, and there could be more unobserved factors that are not being controlled for, causing the possibility for omitted variable bias. For example, veterans may have different unobserved characteristics, such as motivation, work ethic, or social networks, that could affect their wage income that are not being controlled for in the models. Additionally, there could be selection bias, as individuals who choose to join the military may differ systematically from those who do not, in ways that are unable to be captured by the control variables. Therefore, while the results suggest a correlation between veteran status and wage income, they should be interpreted with caution and not as evidence of a causal relationship.

Before moving on I want to return to **Figure 2**. In the figure we saw that veterans earned more than non-veterans overall, but in model 2, we see a significant drop in veteran income compared to non-veteran. This is because in model 2 we included the control variable female, while in figure 2 we only included `log_income` and `Age`. to show this I want to run two regressions, one only include `Age` as a control variable, and then compare it to model 2.

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Table 4: Age Control Regression

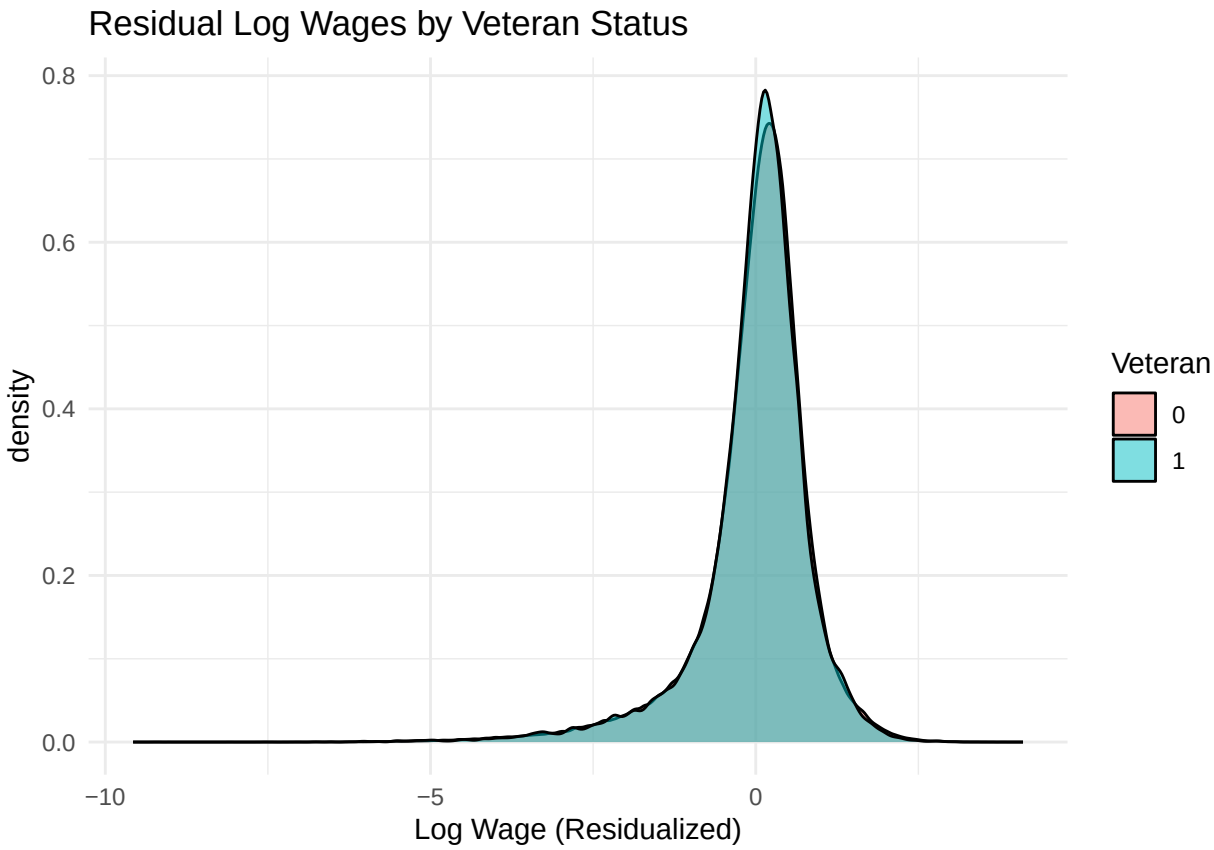
	Dependent variable:	
	Log Wage Income	
	Age	Age + Female
	(1)	(2)
Veteran	0.086***	−0.092***
	(0.007)	(0.007)
Age	0.010***	0.010***
	(0.0001)	(0.0001)
Female		−0.493***
		(0.003)
Constant	10.294***	10.565***
	(0.006)	(0.006)
Observations	500,000	500,000
R^2	0.013	0.068
Adjusted R^2	0.013	0.068
Note:	*p<0.1; **p<0.05; ***p<0.01	

Table 4 illustrates the role of omitted variable bias. When only age is included as a control, the estimated effect of veteran status is positive, suggesting that veterans earn more than non-veteran. However, once gender is controlled for, the coefficient becomes negative and matches the estimate from the second regression.

This change occurs because veteran status is strongly correlated with gender: veterans are much less likely to be female, and female workers earn lower wages on average. When gender is omitted, the veteran coefficient partially captures the effect on gender, biasing the estimate upward. Including gender removes this source of bias, leading to a substantially lower estimate.

This example shows that failing to control for relevant covariates can lead to misleading conclusions about the relationship between veteran status and earnings.

To come back to the density plots, let's show the residuals of the full model to see if there are any differences in the distribution of residuals between veterans and non-veterans, which could indicate that there are still some unobserved factors that are not being controlled for in the regression.



The density plot of the residuals from the full model shows that the distribution of residuals for veterans and non-veterans is similar, with both distributions being centered around zero. However, in the distribution results for veterans, there is a slightly wider spread compared to non-veterans, which could indicate that there are still some unobserved factors that are not being controlled for in the regression that are affecting the wage income of veterans. Additionally, there is some overlap between the two distributions, suggesting that there are some veterans who earn more than some non-veterans, and vice versa, even after controlling for all the variables in the full model. This further supports the idea that there may be unobserved factors at play that are not being captured by the regression.

Discussion of Causal Identification

However, these results should not be interpreted as causal, as veteran status is not randomly assigned, and there could be more unobserved factors that are not being controlled for, causing the possibility for omitted variable bias. For example, veterans may have different unobserved characteristics, such as motivation, work ethic, or social networks, that could affect their wage income that are not being controlled for in the models. Additionally, there could be selection bias, as individuals who choose to join the military may differ systematically from those who do not, in ways that are unable to be captured by the control variables. Therefore, while the results suggest a correlation between veteran status and wage income, they should be interpreted with caution and not as evidence of a causal relationship.

Conclusion

To summarize, this paper studied the effect of veteran status on wage income among those who hold a bachelor's degree. We used data from IPUMS ACS, from 2010 to 2019 controlling for age, disability rating, cognitive and physical difficulty, sex, degree field, occupation, whether they live in a metropolitan area, and survey year fixed effects. In the raw regression, veterans earn up to 14.1% more than non-veterans on average. However, as we start to add more controls, there is a significant decrease in the veteran coefficient. In the full model, veterans earn 2.3% less on average compared to non-veterans, which is statistically significant, suggesting that the wage gap between veterans and non-veterans is substantially reduced after controlling for observable characteristics. However, these results should not be interpreted as causal, as veteran status is not randomly assigned, and there could be more unobserved factors that are not being controlled for, possibly causing omitted variable bias. We cannot observe motivations, work ethic, or social networks of individuals, which could be important factors in explaining the wage gap between veterans and non-veterans. Additionally, there could be selection bias, as individuals who choose to join the military may differ systematically from those who do not, in ways that are unable to be captured by the control variables. Therefore, while the results suggest a correlation between veteran status and wage income, they should be interpreted with caution and not as evidence of a causal relationship.